

1. Write a program to calculate the sum of the ASCII values of all characters in a given string.
Hint: Use the `char` data type and typecasting to `int`.
 2. Create a program that reads the dimensions of a cylinder (radius and height) and calculates its surface area and volume.
 3. Write a program to swap two numbers without using a third variable or arithmetic operators.
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Operators

1. Write a program to check if three numbers can form the sides of a valid triangle.
Hint: Use the triangle inequality property:
 $a+b>c$, $a+c>b$, $b+c>a$
 2. Develop a calculator program that supports addition, subtraction, multiplication, division, and modulus operations. Accept the operator and two operands as input and perform the appropriate operation using a switch statement.
 3. Implement a program to find the largest of three numbers using only relational and logical operators (without using `if-else`).
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Control Statements

1. Write a program to calculate the income tax payable based on the following slab:
 - Income $\leq$ 2,50,000: No Tax
 - Income $>$ 2,50,000 and $\leq$ 5,00,000: 5%
 - Income $>$ 5,00,000 and $\leq$ 10,00,000: 20%
 - Income $>$ 10,00,000: 30%

Use `if-else-if` statements.

2. Develop a program to categorize a given character as:
 - Vowel (a, e, i, o, u)
 - Consonant
 - Digit
 - Special CharacterUse nested `if` statements.
3. Write a program to calculate electricity charges based on units consumed:
 - First 100 units: ₹5/unit
 - Next 100 units: ₹7/unit
 - Beyond 200 units: ₹10/unit

Use `if-else-if` statements.

4. Implement a program to determine if a given year is a "Century Leap Year."
Century Leap Year: A year divisible by 400.

5. **Number Classification:**

Write a program to classify a number as:

- Prime
- Palindrome
- Armstrong

Use nested `if` statements to implement these checks.

6. Create a program that determines if a given 3-digit number is "special." A number is considered special if the sum of the factorial of its digits equals the number itself.